SPT Progress Meeting

November 2026



Omar Mohamed – Practical Work (GBST-BERT)

Institute for Machine Learning

JOHANNES KEPLER
UNIVERSITY LINZ
Altenberger Straße 69
4040 Linz, Austria
jku.at

Recap

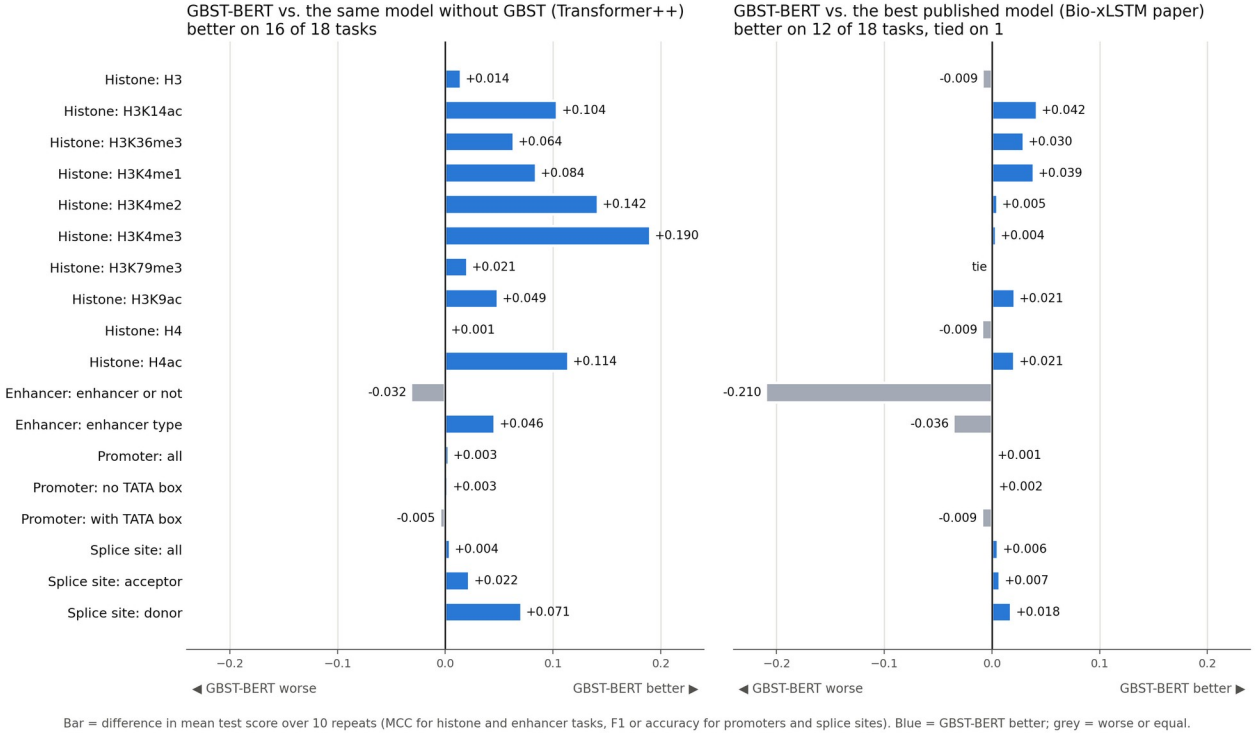
- **Status at the last meeting:** GBST-BERT = learned tokenizer (GBST) + small BERT encoder, 2.1M parameters
  - Pretrained on the human genome (hg38) with the DNA-xLSTM short-context protocol: 10,000 steps
  - Compared with the same model without GBST (Transformer++, 2.0M parameters)
- **Next step decided:** test the model on real biology tasks
  - Nucleotide Transformer benchmark (18 tasks), evaluated exactly like the Bio-xLSTM paper

Progress (1/6): pretraining and benchmark

- **Pretraining (test set):** masked-base accuracy 56.44% vs 56.33% for Transformer++ (loss 0.975 vs 0.977)
- **Nucleotide Transformer benchmark:** 18 tasks × 10 repeats, Bio-xLSTM protocol
  - Histone marks, enhancers, promoters, splice sites
  - Paper's learning rate and batch size per task (Table A5); best validation epoch tested
- **Result:** best of all 7 models on 13 of 18 tasks
  - Better than Transformer++ on 16 of 18 tasks → the learned tokenizer helps
  - Better than the best published model on 12 tasks, tied on 1
  - Weakest task: enhancer or not (−0.21 vs best paper model)

Progress (2/6): GBST-BERT vs other models

- Blue = GBST-BERT better
- Left: same model without GBST
- Right: best model in the Bio-xLSTM paper
- Score = mean test score over 10 repeats



Progress (3/6): results of all 7 models

Task	HyenaDNA	Mamba-PS	Mamba-PH	xLSTM-PS	xLSTM-PH	*ours* GBST-BERT	Transformer++
Histone marks (MCC)							
H3	0.779 ± 0.037	0.799 ± 0.029	0.815 ± 0.048	0.796 ± 0.014	**0.824** ± 0.010	0.815 ± 0.031	0.801 ± 0.029
H3K14ac	0.612 ± 0.065	0.541 ± 0.212	0.631 ± 0.026	0.570 ± 0.008	0.598 ± 0.017	**0.673** ± 0.016	0.569 ± 0.062
H3K36me3	0.613 ± 0.041	0.609 ± 0.109	0.601 ± 0.129	0.588 ± 0.019	0.625 ± 0.010	**0.655** ± 0.035	0.591 ± 0.032
H3K4me1	0.512 ± 0.024	0.488 ± 0.102	0.523 ± 0.039	0.490 ± 0.012	0.526 ± 0.001	**0.565** ± 0.038	0.481 ± 0.037
H3K4me2	0.455 ± 0.095	0.388 ± 0.101	0.487 ± 0.170	0.489 ± 0.024	0.504 ± 0.012	**0.509** ± 0.212	0.367 ± 0.076
H3K4me3	0.549 ± 0.056	0.440 ± 0.202	0.544 ± 0.045	0.520 ± 0.019	0.537 ± 0.012	**0.553** ± 0.180	0.362 ± 0.219
H3K79me3	0.672 ± 0.048	0.676 ± 0.026	**0.697** ± 0.077	0.662 ± 0.011	**0.697** ± 0.007	**0.697** ± 0.132	0.676 ± 0.041
H3K9ac	0.581 ± 0.061	0.604 ± 0.048	0.622 ± 0.030	0.622 ± 0.013	0.627 ± 0.008	**0.648** ± 0.037	0.599 ± 0.046
H4	0.763 ± 0.044	0.789 ± 0.020	0.811 ± 0.022	0.793 ± 0.011	**0.813** ± 0.008	0.804 ± 0.074	0.803 ± 0.022
H4ac	0.564 ± 0.038	0.525 ± 0.240	0.621 ± 0.054	0.558 ± 0.018	0.583 ± 0.014	**0.642** ± 0.037	0.528 ± 0.036
Enhancers (MCC)							
enhancers	0.517 ± 0.117	0.491 ± 0.066	**0.546** ± 0.073	0.375 ± 0.030	0.545 ± 0.024	0.336 ± 0.085	0.368 ± 0.074
enhancers_types	0.386 ± 0.185	0.416 ± 0.095	0.439 ± 0.054	0.444 ± 0.046	**0.466** ± 0.011	0.430 ± 0.167	0.384 ± 0.053
Promoters (F1)							
promoter_all	0.960 ± 0.005	0.967 ± 0.004	0.970 ± 0.004	0.962 ± 0.002	0.967 ± 0.001	**0.971** ± 0.003	0.968 ± 0.006
promoter_no_tata	0.959 ± 0.011	0.968 ± 0.006	0.968 ± 0.010	0.963 ± 0.002	0.970 ± 0.001	**0.972** ± 0.006	0.969 ± 0.004
promoter_tata	0.944 ± 0.040	**0.957** ± 0.015	0.953 ± 0.016	0.948 ± 0.006	0.952 ± 0.005	0.948 ± 0.021	0.953 ± 0.013
Splice sites (accuracy / F1)							
splice_sites_all	0.956 ± 0.011	0.927 ± 0.021	0.940 ± 0.027	0.965 ± 0.006	0.974 ± 0.004	**0.980** ± 0.004	0.975 ± 0.006
splice_sites_acceptors	0.958 ± 0.010	0.936 ± 0.077	0.937 ± 0.033	0.970 ± 0.005	0.953 ± 0.008	**0.977** ± 0.009	0.955 ± 0.049
splice_sites_donors	0.949 ± 0.024	0.874 ± 0.289	0.948 ± 0.025	0.962 ± 0.004	0.951 ± 0.005	**0.980** ± 0.012	0.909 ± 0.117
Tasks with the best mean	**0**	**1**	**2**	**0**	**4**	**13**	**0**

Mean test score over 10 repeats (higher is better); small number = range (best minus worst repeat). Highlighted = best mean for that task (ties counted for each model). First five columns: Bio-xLSTM paper, Table 1.

Progress (4/6): new data – Genomic Benchmarks

- 8 tasks (Grešová et al. 2023) · human, mouse, worm DNA · 5 splits × 10 epochs · test accuracy

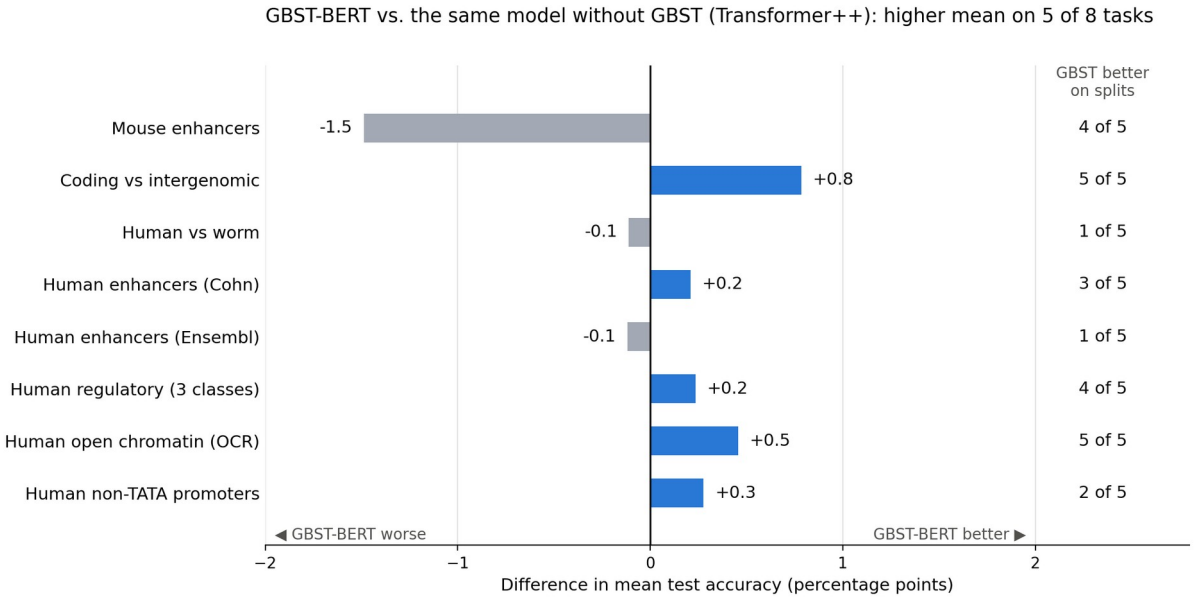
Task (top-1 accuracy)	Mamba-PH 500k	xLSTM-PH 500k	Mamba-PS 500k	xLSTM-PS 500k	*ours* GBST-BERT 2M	*ours* Transformer++ 2M
Mouse enhancers	0.754 ± 0.074	0.780 ± 0.018	**0.793** ± 0.058	0.778 ± 0.007	0.761 ± 0.116	0.776 ± 0.041
Coding vs intergenomic	0.915 ± 0.003	0.931 ± 0.001	0.910 ± 0.003	0.934 ± 0.002	**0.937** ± 0.002	0.930 ± 0.007
Human vs worm	**0.973** ± 0.001	0.965 ± 0.001	0.968 ± 0.002	0.956 ± 0.001	0.971 ± 0.003	0.972 ± 0.002
Human enhancers (Cohn)	**0.747** ± 0.004	0.742 ± 0.005	0.745 ± 0.007	0.734 ± 0.005	0.738 ± 0.019	0.736 ± 0.014
Human enhancers (Ensembl)	0.893 ± 0.008	**0.920** ± 0.001	0.900 ± 0.006	0.902 ± 0.004	0.903 ± 0.008	0.905 ± 0.008
Human regulatory (3 classes)	0.872 ± 0.011	0.872 ± 0.002	0.873 ± 0.007	0.869 ± 0.005	**0.885** ± 0.003	0.883 ± 0.004
Human open chromatin (OCR)	0.828 ± 0.006	0.826 ± 0.002	0.818 ± 0.006	0.800 ± 0.002	**0.841** ± 0.009	0.836 ± 0.008
Human non-TATA promoters	0.946 ± 0.007	0.951 ± 0.004	0.945 ± 0.010	0.949 ± 0.001	**0.955** ± 0.015	0.952 ± 0.004
Tasks with the best mean	**2**	**1**	**1**	**0**	**4**	**0**

Mean test accuracy over 5 train/validation splits (higher is better); small number = range (best minus worst split). Highlighted = best mean.
First four columns: Bio-xLSTM paper, Table A2 (smaller 500k-parameter models).

JKU

Progress (5/6): does GBST help here?

- Higher mean than Transformer++ on 5 of 8 tasks
- Best of all 6 models on 4 of 8
- Gains small (≤ 0.8 points); clearest on coding and open chromatin (5 of 5 splits)
- Mouse enhancers: tiny dataset (1,210 sequences), one bad split
- Paper models are 4× smaller → compare mainly with Transformer++



Progress (6/6): findings and questions

- **What GBST learned:** 98% of its weight on 1-base blocks, so it hardly merges bases
  - The gain probably comes from the extra convolution (local context), not from real "words"
- **Implemented / written:** one clean model file (gbst_bert.py, tested against the training code)
  - Full notebook that re-runs every step on the server; report, poster, slides; all on GitHub
- **Questions :**
  - Is this result enough for the practical work, or should I do more experiments?
  - Learning-rate search per task, or keep the paper's settings for a fair comparison?
  - Longer pretraining (20k steps) or longer sequences (2048)?

JKU

Next steps

- **Before the next meeting :**
  - Small learning-rate / batch-size search on the weakest tasks (enhancers)
  - Check whether GBST merges more bases with longer pretraining or longer sequences
  - Finish the written report with the benchmark results